

Conic graphs

Key Points

Parabolic: $y^2 = 4ax$ is like square root ($y = \sqrt{x}$) but has two values per input

Ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is like a stretched unit circle ($x^2 + y^2 = 1$) by a in x and b in y

Hyperbola: Standard- $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with asymptotes at $y = \pm \frac{b}{a}x$ and Rectangular- $xy = c^2$ is like reciprocal ($y = \frac{1}{x}$) with asymptotes $x = 0$ and $y = 0$

Basic Skills

Q1. a) Sketch the graph of $y = 2\sqrt{x}$

b) Sketch the graph of $y^2 = 4x$

Q2. a) Draw the unit circle $x^2 + y^2 = 1$

b) Determine the scale factors in both the x -direction and y -direction in the transform to the graph $\frac{x^2}{4} + \frac{y^2}{9} = 1$

c) Hence sketch the graph of $\frac{x^2}{4} + \frac{y^2}{9} = 1$

Q3. a) State the asymptotes of the graph $\frac{x^2}{4} - \frac{y^2}{9} = 1$

b) Sketch the graph of $\frac{x^2}{4} - \frac{y^2}{9} = 1$

Q4. Draw the graph of $xy = 4$

Competency Skills

Q5. The point $A=(2,2)$ and Line L is $x = 6$

a) Write down the distance of any point $P=(x,y)$ away from A (i.e. AP)

b) Write down the distance of any point $P=(x,y)$ away from L (i.e. LP)

c) Hence find an equation for the locus of all points such that the distance from A is twice the distance from L

Q6. Draw the following graphs

a) $\frac{(x+1)^2}{16} + \frac{(y-2)^2}{25} = 1$ b) $xy = -8$ c) $\frac{x^2}{5} - \frac{(2y-5)^2}{25} = 1$

Q7. The graph $\frac{x^2}{16} - y^2 = 1$ is stretched by scale factor 5 in the x -direction and then reflected in the x -axis state the equation of the new graph

Q8. A transformed parabolic curve passes through the point $(-3,0)$ state a possible equation for the curve

Advanced Skills

Q9. Reflect the graph of $\frac{x^2}{5} - \frac{y^2}{2} = 2$ in the line $y = x$ stating the new equation and sketching the graph

Q10. The parabolic curve $y^2 = 2x$ is translated by $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$ and then stretched by scale factor 0.5 in the direction parallel to the x -axis. Write down the new equation

Q11. The rectangular hyperbola $xy = 8$ and standard hyperbola $\frac{x^2}{4} - y^2 = 1$ intersect, find the x -ordinates of intersection

Q12. By using the difference of two squares explain why the standard hyperbola has asymptotes $y = \pm \frac{b}{a}x$

Q13. Draw the graph $y^2 + (3x - 2)^2 = 25$

Extension

Q14. Determine the conditions required for an ellipse to intersect a standard hyperbola